Lopcombe Corner — Does the Level at the Start of Winter Matter?

Winter peak vs Oct–Mar rainfall with and without the 1 October groundwater level · 59 winters 1963/64–2024/25 · levels: full EA record (manual dips and daily logger) · rainfall: HadUK-Grid 1 km, borehole cell · updated 20 Oct 2026

Bottom line. Yes. Adding the 1 October level to Oct–Mar rainfall raises the share of winter-peak variation explained from 0.69 to **0.79** and cuts the typical error from 3.4 m to 2.8 m. Each extra metre of starting level adds about **4.5 m** to the peak ($p < 0.001$). Because the well drains back to almost the same level every autumn (78.45 ± 0.43 m), rainfall is still the main driver — but a high start makes a flood noticeably more likely.

This replaces the version issued earlier on 20 Oct 2026, which used 25 winters and logger-era peaks that were later found to be too low in several years. With the full EA record the effect is stronger and is now tested over all winters back to 1963/64.

1. What the start level is

- The level on 1 October, interpolated between the nearest readings (within 21 days), before winter recharge begins. It matches the autumn recession low used earlier to within a few centimetres.
- Across 59 winters it averaged 78.45 m (standard deviation 0.43 m), from 77.15 m (1976/77, after the 1976 drought) to 79.39 m (2008/09).
- It is unrelated to that winter's rainfall ($r = -0.03$), so it adds independent information. On its own it correlates with the peak at $r = 0.29$.

2. How the finding has developed

Analysis	Data	Without start	With start	Start effect
v1, v2 (7 Sep)	Broughton Oct–Jan / Oct–Dec, first-pass peaks	0.52 / 0.43	0.56 / 0.49	weak, not significant
v3 (7 Sep)	Broughton Oct–Mar, corrected peaks, 25 winters	0.74	0.83	≈ +5 m per m
20 Oct, first issue	HadUK Oct–Mar, peaks later found too low, 25 winters	0.80	0.85	+3.6 m per m, p 0.02
This report	HadUK Oct–Mar, full record, logger era (26)	0.76	0.86	+5.0 m per m, p 0.001
This report	**HadUK Oct–Mar, full record, all winters (59)**	**0.69**	**0.79**	**+4.5 m per m, p < 0.001**

Columns 3–4 are R^2 (share of variation in winter peak explained).

- The early analyses saw little effect because their rainfall window missed Feb–Mar recharge and several peaks were truncated. Fixing those made the effect clear.
- This morning's version understated it because the peaks it used were too low in several winters that had started high — 2024/25 (start 79.04 m; peak 95.04 m, not 90.31 m) and 2012/13 (start 79.04 m; peak 95.73 m, not 94.42 m). With the full record the effect is larger and much more certain.
- Using Oct–Jan rain instead: R^2 0.59 → 0.71, start effect +4.9 m per m — the same picture.

3. Current model (all 59 winters)

Model	Term	Coefficient	Std error	p-value
Rain only	Intercept	69.133	1.791	< 0.001
	Oct–Mar rain (per mm)	0.039	0.003	< 0.001
Rain + start level	Intercept	-284.545	70.062	< 0.001
	Oct–Mar rain (per mm)	0.040	0.003	< 0.001
	Start level (per m)	4.506	0.892	< 0.001

Fit: peak ≈ -284.5 + 0.040 × Oct–Mar rain + 4.51 × 1 October level. F-test for adding the start level: $F = 25.5$, $p < 0.001$. Partial correlation of start level with the peak after allowing for rain: $r = 0.56$. 95% range for the start effect: about 2.7–6.3 m per m.

In practical terms: a start one standard deviation (0.43 m) above average raises the expected peak by about 1.9 m. By comparison, 100 mm more Oct–Mar rain raises it by about 4.0 m, and one standard deviation of Oct–Mar rain (127 mm) by about 5.0 m. So rainfall matters roughly two to three times as much, but a high start is worth about 50 mm of extra winter rain.

4. Where the start level helps — and where it doesn't

- **Helps most:** 1976/77 (start 77.15 m; residual –6.4 → –0.6 m), 2008/09 (79.39 m; +7.4 → +3.2 m), 1997/98 (77.64 m; –4.8 → –1.2 m), 2005/06 (77.70 m; –5.1 → –1.7 m), 1974/75 (79.08 m; +5.6 → +2.7 m) and 2024/25 (79.04 m; +6.2 → +3.6 m) — the very low and very high starts.
- **Makes it worse:** 1989/90 and 2019/20 (low starts, but a very wet February drove a high peak), and 2023/24 and 2007/08 (high starts that peaked lower than the model expects).
- **Still unexplained:** 1965/66 (+8.7 m), 1981/82 (+5.8 m), 2018/19 (–5.2 m), 1968/69 (+4.7 m) and 1999/2000 (+4.3 m) — pointing to rainfall timing within the winter as the next missing factor.
- **Flood-threshold calls:** both models flag four of the six flood winters with rainfall data and give no false alarms. Rain alone misses 1994/95 and 2002/03; with the start level it catches 1994/95 but misses 2019/20 (low start, intense February rain). 2002/03 is missed by both.

5. Caveats and next steps

- Before 1999 the start level and peak come from roughly weekly manual dips, so both can be slightly smoothed.
- Winter 2025/26 (peak 99.02 m) is not included because HadUK-Grid rainfall currently ends in December 2025.
- Three winters have no reading close enough to 1 October (1973/74, 1986/87, 1995/96) and are left out.
- Next: add a rainfall-timing term (for example February rain or the wettest 30 days) using the monthly HadUK-Grid data, and use the start level in the month-by-month flood-probability model.

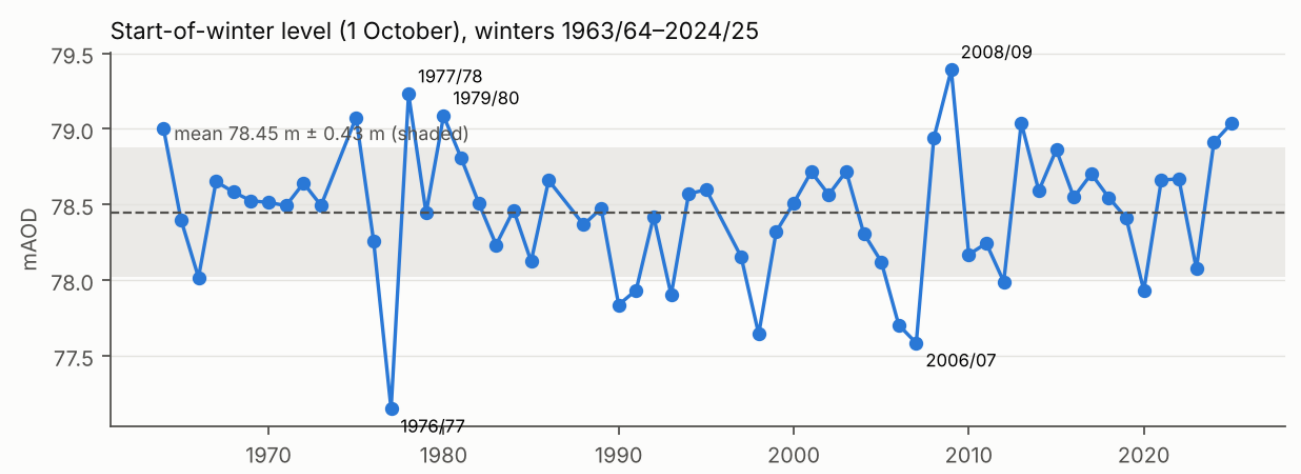
Reproduce: `python scripts/winter_summary.py`, then `python scripts/start_level_analysis.py` (all winters) and `--first-winter 2000 --tag _logger` (logger era). Outputs in output/haduk/.

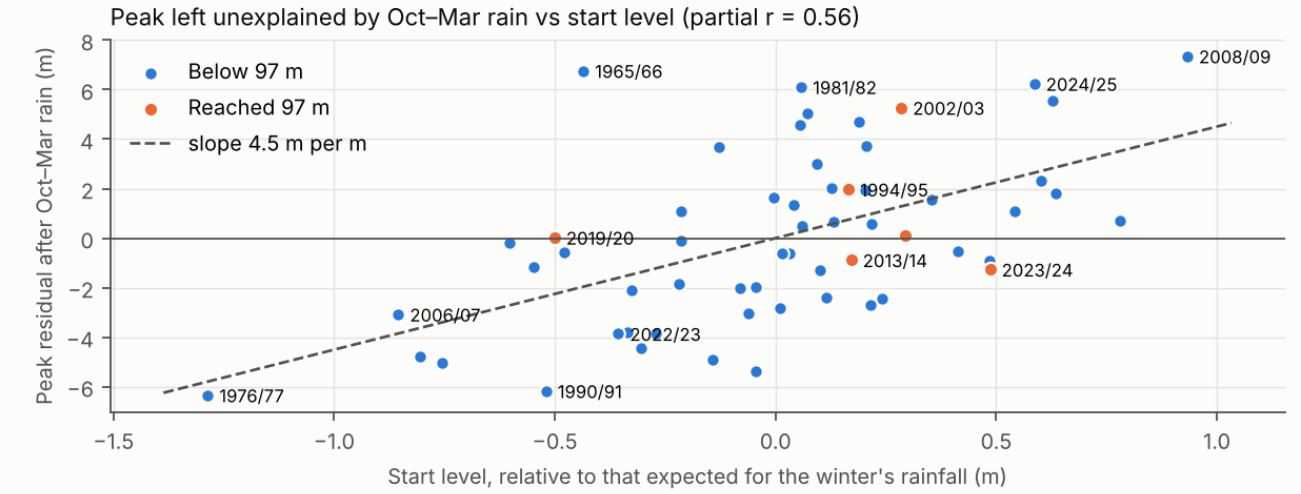
Appendix — winter by winter (all winters)

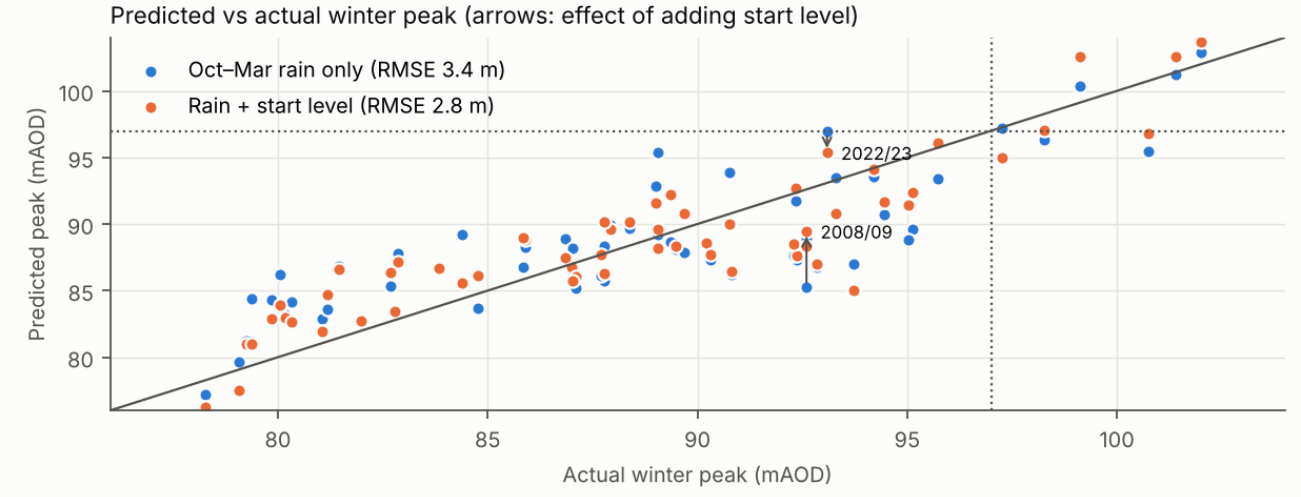
Winter	Oct–Mar mm	Start m	Peak m	Pred: rain	Pred: + start	Resid: rain	Resid: + start
1963/64	371	79.00	84.76	83.7	86.1	+1.1	-1.4
1964/65	359	78.40	80.18	83.2	82.9	-3.0	-2.8
1965/66	454	78.02	93.73	87.0	85.0	+6.8	+8.7
1966/67	550	78.65	94.44	90.7	91.7	+3.7	+2.8
1967/68	423	78.58	87.78	85.8	86.3	+2.0	+1.5
1968/69	464	78.53	92.38	87.3	87.7	+5.0	+4.7
1969/70	443	78.51	87.01	86.5	86.8	+0.5	+0.2
1970/71	484	78.49	89.48	88.1	88.3	+1.4	+1.2
1971/72	471	78.64	92.30	87.6	88.5	+4.7	+3.8
1972/73	343	78.49	81.97	82.6	82.7	-0.6	-0.8
1974/75	521	79.08	95.12	89.6	92.4	+5.5	+2.7
1975/76	205	78.26	78.26	77.2	76.2	+1.1	+2.0
1976/77	669	77.15	89.05	95.4	89.6	-6.3	-0.6
1977/78	498	79.23	89.37	88.7	92.2	+0.7	-2.8
1978/79	495	78.44	90.22	88.6	88.5	+1.6	+1.7
1979/80	478	79.09	89.69	87.9	90.8	+1.8	-1.1
1980/81	434	78.81	87.71	86.2	87.7	+1.6	-0.0
1981/82	448	78.51	92.84	86.7	87.0	+6.1	+5.8
1982/83	511	78.23	89.07	89.2	88.2	-0.1	+0.9
1983/84	447	78.46	83.85	86.7	86.7	-2.8	-2.9
1984/85	504	78.12	86.84	88.9	87.5	-2.1	-0.6
1985/86	408	78.66	87.11	85.2	86.1	+2.0	+1.0
1987/88	530	78.37	87.92	89.9	89.6	-2.0	-1.6
1988/89	364	78.47	82.78	83.4	83.5	-0.6	-0.7
1989/90	620	77.84	93.30	93.5	90.8	-0.2	+2.5
1990/91	436	77.94	80.06	86.2	83.9	-6.2	-3.8
1991/92	308	78.42	79.24	81.2	81.0	-2.0	-1.8
1992/93	486	77.90	87.02	88.2	85.7	-1.2	+1.3
1993/94	622	78.57	94.20	93.5	94.1	+0.7	+0.1
1994/95	**692**	**78.60**	**98.26**	**96.3**	**97.0**	**+2.0**	**+1.2**
1996/97	386	78.15	79.84	84.3	82.9	-4.5	-3.1
1997/98	511	77.64	84.38	89.2	85.5	-4.8	-1.2

Winter	Oct–Mar mm	Start m	Peak m	Pred: rain	Pred: + start	Resid: rain	Resid: + start
1998/99	504	78.32	92.59	88.9	88.3	+3.7	+4.3
1999/00	436	78.51	90.82	86.2	86.5	+4.6	+4.3
2000/01	**819**	**78.72**	**101.39**	**101.3**	**102.6**	**+0.1**	**-1.2**
2001/02	488	78.57	85.90	88.3	88.8	-2.4	-2.9
2002/03	**671**	**78.72**	**100.75**	**95.5**	**96.8**	**+5.3**	**+4.0**
2003/04	475	78.31	82.85	87.8	87.1	-4.9	-4.3
2004/05	382	78.12	80.32	84.2	82.6	-3.8	-2.3
2005/06	390	77.70	79.37	84.4	81.0	-5.1	-1.7
2006/07	630	77.58	90.76	93.8	90.0	-3.1	+0.8
2007/08	449	78.94	85.85	86.8	88.9	-0.9	-3.1
2008/09	411	79.39	92.60	85.2	89.5	+7.3	+3.1
2009/10	604	78.17	89.00	92.8	91.6	-3.8	-2.6
2010/11	351	78.24	81.06	82.9	81.9	-1.9	-0.9
2011/12	268	77.99	79.08	79.7	77.5	-0.6	+1.6
2012/13	619	79.04	95.73	93.4	96.1	+2.3	-0.4
2013/14	**860**	**78.59**	**102.01**	**102.9**	**103.7**	**-0.9**	**-1.7**
2014/15	489	78.86	87.78	88.3	90.2	-0.6	-2.4
2015/16	524	78.55	88.39	89.7	90.2	-1.3	-1.8
2016/17	369	78.70	81.18	83.6	84.7	-2.4	-3.5
2017/18	464	78.55	90.31	87.3	87.8	+3.0	+2.6
2018/19	451	78.41	81.46	86.8	86.6	-5.4	-5.2
2019/20	**716**	**77.93**	**97.25**	**97.2**	**95.0**	**+0.0**	**+2.3**
2020/21	576	78.66	92.33	91.8	92.7	+0.6	-0.4
2021/22	414	78.67	82.68	85.4	86.3	-2.7	-3.7
2022/23	709	78.08	93.09	97.0	95.3	-3.9	-2.3
2023/24	**796**	**78.91**	**99.11**	**100.4**	**102.6**	**-1.3**	**-3.5**
2024/25	501	79.04	95.04	88.8	91.5	+6.2	+3.6

Bold rows reached 97 m. Residual = actual peak minus predicted.


Start-of-winter level (1 October), winters 1963/64–2024/25
mAOD
mean 78.45 m ± 0.43 m (shaded)
1977/78
1979/80
2008/09
2006/07
1976/77


Peak left unexplained by Oct–Mar rain vs start level (partial r = 0.56)
Peak residual after Oct–Mar rain (m)
Below 97 m
Reached 97 m
slope 4.5 m per m
1965/66
1981/82
2002/03
2024/25
2008/09
1994/95
2019/20
2013/14
2023/24
2006/07
2022/23
1976/77
1990/91
Start level, relative to that expected for the winter's rainfall (m)


Predicted vs actual winter peak (arrows: effect of adding start level)
Predicted peak (mAOD)
Oct–Mar rain only (RMSE 3.4 m)
Rain + start level (RMSE 2.8 m)
2022/23
2008/09
Actual winter peak (mAOD)